

PRODHI MANISHA

AI Engineer, Strategist & Solutions Architect

 mprodhi@gmail.com |  +55 (31) 97179-0472
 linkedin.com/in/prodhi-manisha |  github.com/prodm93
 huggingface.co/prodm93
 Remote; flexible working hours

PROFESSIONAL SUMMARY

Biomedical scientist turned AI solutions architect with 4+ years of specialised experience developing human-centred AI/ML systems at the intersection of biotech and AI. Founder of Confluente Ltda, an AI consultancy and solutions architecture firm focused on ethical, scalable ML solutions. Proven track record building end-to-end RAG applications, fine-tuning open-source models (LLaMA, Vicuna), ML pipelines for clinical and pharmaceutical data, therapeutic AI chatbots, and biomedical/healthcare knowledge synthesis systems. I combine deep biomedical research experience and scientific methodology with cutting-edge AI/ML expertise to create meaningful healthtech innovations, and make AI work for humans instead of the other way round.

EXPERIENCE

Founder & AI Solutions Architect

Confluente Ltda | Oct 2025 – Present

AI consultancy and solutions architecture firm specialising in ethical, effective and human-centred AI systems.

Lead AI/ML Engineering Contractor

Healthcare & Biotech Clients | Apr 2023 – Present

- Pharmaceutical Analysis (Danaher via NILG.ai): Developed computational pipeline for drug compound classification and complex clinical trial data processing
- AI Scientific Literature & Citations Insights Pipeline (NILG.ai): Constructed innovation analysis workflow for computational chemistry companies embedded in unstructured Pubmed literature/citations data
- LLM Therapeutic Chatbot (NILG.ai): Built sophisticated conversation flow management for psychological therapy assistance involving dynamic topic exploration
- In-Place DPO Implementation: Architected novel workflow applying Direct Preference Optimisation principles to few-shot prompting. Programmatically processed real-time collaborative editing data (highlights, strikeouts, multi-user edits) from Notion database, automatically generating structured JSON paired datasets (positive/negative examples with improvement annotations) to enhance content generation quality
- AI Game Generation and Counselling Chatbot (Funify): Architected personalised game generation and counselling mediation system via WhatsApp chatbot
- RAG Q&A Systems: Implemented zero-cost document chat and agentic graphRAG query systems, extendable to healthcare data and documentation
- Antiviral Drug Discovery: Created antiviral peptide ML classification and scoring system

NLP Data Scientist

Sagewrite ehf | Apr 2022 – Apr 2023

Built text extraction pipelines for academic literature, scientific text generation models, and cloud-deployed research applications.

TECHNICAL SKILLS

Core ML Engineering Stack

LLM Development & Deployment: RAG & graphRAG pipelines, agentic workflows (LangGraph, OpenAI SDK, CrewAI), MCP servers, LangChain (custom backend refinement), prompt engineering, fine-tuning (LoRA/qLoRA), model quantisation (GGUF, AWQ), serving local LLMs, protein language models

Frameworks: HuggingFace Transformers, PyTorch, TensorFlow, Scikit-learn

MLOps, API Workflows & Cloud: AWS (EC2, S3), GCP, Azure, Docker, Git, API integration & development (FastAPI, Flask), Streamlit, Render, n8n automation

Data Engineering: Web scraping (Selenium/BeautifulSoup), big dataset visualisation (Plotly, Matplotlib), Pandas, biomedical data analysis, clinical statistics, knowledge graphs

Healthcare AI & Bioinformatics: Biomedical/clinical/healthcare knowledge synthesis, pharmaceutical informatics, bioinformatics pipelines, therapeutic chatbots

Languages: Python, SQL, R

LANGUAGES

English: Native

French: Conversational

Icelandic: Intermediate

Portuguese: Basic

Spanish: Basic

CORE EXPERTISE

- Human-centred AI design
- Therapeutic AI systems
- Clinical data analysis
- Biomedical informatics
- Ethical AI development
- Scientific methodology

Doctoral Researcher

Amsterdam UMC - Glycobiology & Immunology Lab | Dec 2020 – Dec 2021

Cancer immunotherapy research: analysed sialidases for tumour targeting, designed pH-responsive therapeutic liposomes, and contributed to framework development for next-generation cancer immunotherapies.

Graduate Research Fellow

CIC BioGUNE | Jan 2020 – Oct 2020

Breakthrough Discovery: Identified novel Siglec-15 glycoform mechanism suppressing immune responses in cancer. Experimentally proved that glycosylation removal abrogates T cell binding, thus establishing a new therapeutic target for cancer treatment.

Junior Research Scholar

Thayer School of Engineering at Dartmouth | Sep 2015 – Jun 2017

Engineered novel yeast high-throughput screening/display systems for directed evolution and anti-B7H6 CAR-T constructs for cancer immunotherapy.

EDUCATION

MSc Applied Data Science & AI | Open Institute of Technology | 2025-26
(Expected)

MSc Molecular Biology & Biomedicine | University of the Basque Country |
2019-2020 | 9.15/10.0

BSc(Hons) Combined STEM | The Open University, UK | 2018-2019 | First Class
Honours

Engineering Sciences (Bioengineering) | Dartmouth College, USA | 2013-17 | 193
ECTS credits

KEY ACHIEVEMENTS

- Scientific Breakthrough: Discovered glycoform-dependent Siglec-15 immune suppression mechanism for therapeutic targeting
- Published: Nature Communications co-author (2023) on SIGLEC-15 research (<https://doi.org/10.1038/s41467-023-39119-8>)
- Marie Skłodowska-Curie Actions and Leiden Institute for Chemical Immunology ESR grant recipient
- Perfect client satisfaction rating across 23+ Upwork projects
- Healthcare Innovation: Biomedical, healthtech & therapeutic AI systems with measurable patient outcome and engagement improvements

REMOTE WORK & COLLABORATION

- Autonomy: Successfully managed 23+ freelance projects remotely with clients spread across 5+ time zones
- Startup Pace: Startup-native ML training; delivered production MVPs in 2-4 week sprints; iterated based on supervisor and customer feedback
- Adaptability: Transitioned from biomedical research to ML engineering on own initiative; mastered modern stack in 18 months
- Positivity: 100% of client reviews highlight "great communication" and "dedicated collaborator"--a people person on a mission to break the 'recluse scientist/programmer' stereotype

PRODHI MANISHA

Automation Specialist & GenAI Engineer

✉ mprodhi@gmail.com |
☎ +55 (31) 97179-0472
🔗 linkedin.com/in/prodhi-manisha
🏠 github.com/prodm93
🐙 huggingface.co/prodm93
📍 Remote; flexible working hours

TECHNICAL SKILLS

AI/ML Development

LLM applications and fine-tuning (LoRA, QLoRA, PEFT), RAG systems, prompt engineering, PyTorch, TensorFlow, HuggingFace, LangChain, OpenAI & Anthropic APIs

Automation & Infrastructure

n8n workflows, agentic workflows (OpenAI SDK, LangGraph, AutoGen, CrewAI) and tool incorporation, MCP servers, (agentic) graphRAG (Neo4j, Graphiti), Airtable integrations, API development, Docker, AWS, GCP, Azure, CI/CD pipelines, Claude Code, Codex, Cursor, GitHub Copilot

Development

Python, SQL, R, Git, API design, database architecture

Data Science

Pandas, Scikit-learn, data pipeline design, computational analysis, statistical modelling

LANGUAGES

English: Native
French: Conversational
Icelandic: Intermediate
Portuguese: Basic
Spanish: Basic

CORE EXPERTISE

- End-to-end automation architecture
- Custom GenAI model development
- Data pipeline optimisation
- API integration & development
- Cloud infrastructure deployment
- Scientific methodology

PROFESSIONAL SUMMARY

Automation specialist and GenAI engineer with 4+ years of hands-on experience building scalable AI systems and complex automation workflows for multiple corporate clients via long-term contractual engagements. Founder of Confluent Ltda, specialising in end-to-end technical solutions from infrastructure design to deployment. Expert in LLM applications, prompt engineering, and n8n automation pipelines. I combine deep scientific methodology with cutting-edge AI/ML expertise to create robust systems that solve real-world problems, and help AI become your supercharging sidekick beyond the buzzwords and hype.

EXPERIENCE

Founder & Lead Automation Architect

Confluent Ltda | Oct 2025 – Present

Leading technical firm for automation solutions architecture and GenAI system development. Building scalable AI infrastructure, custom automation and agentic workflows for clients.

GenAI Developer & Automation Engineer

Contract: Berkshire Partners · Danaher (via NILG.ai) · Rockridge.ai · Others | Apr 2023 – Present

- Multi-Platform Content Automation Ecosystem: Architected comprehensive system integrating n8n workflows, Airtable databases, APIs for Claude and multiple LLM providers along with social media APIs, automating 60% of content production pipeline from generation to publication; saved 4 hours/week per workflow
- AI-Powered Visual Content Generation: Built automated carousel and reels generation system using OpenAI's image/video models with brand guideline enforcement, along with automatic brand application and AWS S3 integration
- In-Place DPO Implementation: Architected novel workflow applying Direct Preference Optimisation principles to few-shot prompting. Programmatically processed real-time collaborative editing data (highlights, strikeouts, multi-user edits) from Notion database, automatically generating structured JSON paired datasets (positive/negative examples with improvement annotations) to enhance content generation quality. A/B testing demonstrated ~30% overall quality improvement without fine-tuning overhead.
- Intelligent Content Calendar System: Developed Notion-to-Airtable-to-RecurPost automated publishing pipeline with pre-publish verification and last-minute update detection
- Real-Time Analytics Infrastructure: Created comprehensive multi-platform performance tracking system with automated KPI reporting, engagement correlation analysis, and content optimisation feedback loops
- Personalised Game Generation Pipeline (Funify): Architected LLM backend of complete AI-driven system generating custom games via WhatsApp chatbot
- Enterprise Content Automation: Built comprehensive n8n workflow ecosystem integrating social media APIs, Airtable databases, and AI content generation
- Private Investment Intelligence System: Developed secure RAG-based LLM system for Berkshire Partners processing confidential stakeholder transcripts to generate business forecasts and investment opportunities
- Pharmaceutical Data Pipeline (Danaher via NILG.ai): Created computational analysis pipeline for drug compound classification and clinical trial data processing with AI/ML
- LLM Model Optimisation: Fine-tuned and quantised multiple LLMs (LLaMA, BERT, GPT variants) using LoRA/QLoRA techniques, reducing inference costs significantly
- Zero-Cost RAG Systems: Implemented document chat systems using open-source LLMs and vector databases, providing enterprise-grade functionality without API dependencies

Data Science & Machine Learning Trainee/Contractor

Sagewrite ehf | Apr 2022 – Apr 2023

Trained in and later contracted for academic AI application technical projects. Built scalable text processing pipelines and supervised ML model development.

Research Scientist

Amsterdam UMC & CIC BioGUNE | Jan 2020-Dec 2021

Directed breakthrough biomedical research leading to novel therapeutic discovery, published in Nature Communications. Led technical projects from hypothesis to production-ready systems.

EDUCATION

MSc Applied Data Science & AI | Open Institute of Technology | 2025-26 (Expected)

MSc Molecular Biology & Biomedicine | University of the Basque Country | 2019-2020 | 9.15/10.0

BSc(Hons) Combined STEM | The Open University, UK | 2018-2019 | First Class Honours

Engineering Sciences (Bioengineering) | Dartmouth College, USA | 2013-2017 | 193 ECTS credits

KEY ACHIEVEMENTS

- Technical Innovation: Developed novel automation architectures and complex orchestrations reducing manual processing time by ~60% across multiple client projects
- Perfect Client Delivery: 100% satisfaction rating on Upwork across 23+ technical projects
- Novel Cancer Therapeutic Discovery: Led research published in Nature Communications establishing new immunotherapy target for cancer treatment

PRODHI MANISHA

AI Product Manager, Strategist & Solutions Architect

 mprodhi@gmail.com |  +55 (31) 97179-0472
 linkedin.com/in/prodhi-manisha |  github.com/prodm93
 Remote; flexible working hours

PROFESSIONAL SUMMARY

AI Product Manager and Solutions Architect with 4 years building scalable AI systems that drive business impact. Founder of Confluente Ltda, specialising in automation solutions architecture and human-centred AI product development. Proven track record delivering end-to-end AI solutions from strategy to implementation, with expertise in workflow automation, n8n pipelines, and cross-functional team leadership. I bridge the gap between cutting-edge AI capabilities and practical business applications, ensuring technology serves strategic objectives.

EXPERIENCE

Founder & Owner

Confluente Ltda | Oct 2025 – Present

Leading AI company for automation solutions architecture and strategic AI product development. Building scalable AI systems that deliver measurable ROI for clients across multiple industries.

AI Product Lead & Solutions Architect

Various Clients (Freelance) | Apr 2023 – Present

- **Product Strategy:** Led end-to-end AI product development from ideation to deployment, delivering personalised game generation system via WhatsApp (Funify)
- **Automation Architecture:** Built automated social media pipelines; integrated analytics dashboards; and comprehensive content generation ecosystem with n8n workflows. Implemented novel in-place DPO methodology, programmatically cleaning messy collaborative Notion database to create structured positive/negative example datasets for few-shot prompting. A/B testing showed ~30% quality improvement.
- **Strategic Private LLM System:** Architected secure, private AI system for Berkshire Partners, processing confidential stakeholder interview transcripts to generate business forecasts, company predictions, and investment opportunities while ensuring enterprise-grade data security and compliance
- **Enterprise AI Solutions:** Developed pharmaceutical analysis pipeline for Danaher, delivering computational insights for drug compound classification and clinical trial processing
- **Technical Product Management:** Managed AI model lifecycle from research to production, including LLM fine-tuning, evaluation frameworks, and deployment strategies

AI Research & Development Lead

Sagewrite ehf | Apr 2022 – Apr 2023

Led technical research initiatives for academic AI applications. Built scalable text processing pipelines and supervised ML model development for scientific content generation.

Research Scientist & Innovation Lead

Amsterdam UMC & CIC BioGUNE | Jan 2020-Dec 2021

Directed breakthrough research leading to novel therapeutic target discovery. Published findings in Nature Communications. Demonstrated ability to lead complex technical projects from hypothesis to biomedical industry-relevant outcomes.

EDUCATION

MSc Applied Data Science & AI | Open Institute of Technology | 2025-26 (Expected)
MSc Molecular Biology & Biomedicine | University of the Basque Country | 2019-2020 | 9.15/10.0
BSc(Hons) Combined STEM | The Open University, UK | 2018-2019 | First Class Honours
Engineering Sciences (Bioengineering) | Dartmouth College, USA | 2013-17 | 193 ECTS credits

KEY ACHIEVEMENTS

- **Business Impact:** 100% client satisfaction across 23+ freelance AI-focused projects
- **Product Innovation:** Delivered AI products that increased user engagement by 40+% and automated ~60% of content workflows (human time saved: ~4 hours a week per workflow)
- **Technical Leadership:** Published Nature Communications research, demonstrating ability to drive complex projects to completion
- **Strategic Vision:** Successfully transitioned from research to commercial AI applications, bridging academic rigour with business value

TECHNICAL SKILLS

AI/ML Platforms

LLM fine-tuning, RAG systems, PyTorch, TensorFlow, HuggingFace, LangChain, OpenAI and Anthropic APIs

Product & Strategy

Product roadmapping, user research, A/B testing, analytics dashboards, KPI tracking, Agile workflows, Jira/Asana

Automation Tools

n8n workflows, Airtable integrations, API development

Development

Python, SQL, R, Git, Docker, AWS, GCP, Azure, Render dashboards

CORE COMPETENCIES

- AI Product Management
- Solutions Architecture
- Automation Strategy
- Cross-functional Leadership
- Technical Product Strategy
- Human-centred Design

LANGUAGES

English: Native
French: Conversational
Icelandic: Intermediate
Portuguese: Basic
Spanish: Basic

PRODHI MANISHA

Automation Specialist & GenAI Engineer

✉ mprodhi@gmail.com |
☎ +55 (31) 97179-0472
🔗 linkedin.com/in/prodhi-manisha
🐙 github.com/prodm93
🤖 huggingface.co/prodm93
📍 Remote; flexible working hours

TECHNICAL SKILLS

AI/ML Development

LLM applications and fine-tuning (LoRA, QLoRA, PEFT), RAG systems, prompt engineering, PyTorch, TensorFlow, HuggingFace, LangChain, OpenAI & Anthropic APIs

Automation & Infrastructure

n8n workflows, agentic workflows (OpenAI SDK, LangGraph, AutoGen, CrewAI) and tool incorporation, MCP servers, (agentic) graphRAG (Neo4j, Graphiti), Airtable integrations, API development, Docker, AWS, GCP, Azure, CI/CD pipelines, Claude Code, Codex, Cursor, GitHub Copilot

Development

Python, SQL, R, Git, API design, database architecture

Data Science

Pandas, Scikit-learn, data pipeline design, computational analysis, statistical modelling

LANGUAGES

English: Native
French: Conversational
Icelandic: Intermediate
Portuguese: Basic
Spanish: Basic

CORE EXPERTISE

- End-to-end automation architecture
- Custom GenAI model development
- Data pipeline optimisation
- API integration & development
- Cloud infrastructure deployment
- Scientific methodology

PROFESSIONAL SUMMARY

Automation specialist and GenAI engineer with 4+ years of hands-on experience building scalable AI systems and complex automation workflows. Founder of Confluente Ltda, specialising in end-to-end technical solutions from infrastructure design to deployment. Expert in LLM applications, prompt engineering, agentic and automation pipelines. I combine deep scientific methodology with cutting-edge AI/ML expertise to create robust systems that solve real-world problems, and help AI become your supercharging sidekick beyond the buzzwords and hype.

EXPERIENCE

Founder & Lead Automation Architect

Confluente Ltda | Oct 2025 – Present

Leading technical firm for automation solutions architecture and GenAI system development. Building scalable AI infrastructure, custom automation and agentic workflows for clients.

GenAI Developer & Automation Engineer

Contract (Various Clients) | Apr 2023 – Present

- Multi-Platform Content Automation Ecosystem: Architected comprehensive system integrating n8n workflows, Airtable databases, Claude API, and social media APIs, automating 60% of content production pipeline from generation to publication
- AI-Powered Visual Content Generation: Built automated carousel and reels generation system using OpenAI's image/video models with brand guideline enforcement, along with automatic brand application and AWS S3 integration
- In-Place DPO Implementation: Architected novel workflow applying Direct Preference Optimisation principles to few-shot prompting. Programmatically processed real-time collaborative editing data (highlights, strikeouts, multi-user edits) from Notion database, automatically generating structured JSON paired datasets (positive/negative examples with improvement annotations) to enhance content generation quality. A/B testing demonstrated ~30% overall quality improvement without fine-tuning overhead.
- Intelligent Content Calendar System: Developed Notion-to-Airtable-to-RecurPost automated publishing pipeline with pre-publish verification and last-minute update detection
- Real-Time Analytics Infrastructure: Created comprehensive multi-platform performance tracking system with automated KPI reporting, engagement correlation analysis, and content optimisation feedback loops
- Personalised Game Generation Pipeline (Funify): Architected LLM backend of complete AI-driven system generating custom games via WhatsApp chatbot
- Enterprise Content Automation: Built comprehensive n8n workflow ecosystem integrating social media APIs, Airtable databases, and AI content generation
- Private Investment Intelligence System: Developed secure RAG-based LLM system for Berkshire Partners processing confidential stakeholder transcripts to generate business forecasts and investment opportunities
- Pharmaceutical Data Pipeline (Danaher via NILG.ai): Created computational analysis pipeline for drug compound classification and clinical trial data processing with AI/ML
- LLM Model Optimisation: Fine-tuned and quantised multiple LLMs (LLaMA, BERT, GPT variants) using LoRA/QLoRA techniques, reducing inference costs significantly
- Zero-Cost RAG Systems: Implemented document chat systems using open-source LLMs and vector databases, providing enterprise-grade functionality without API dependencies

Data Science & Machine Learning Trainee/Contractor

Sagewrite ehf | Apr 2022 – Apr 2023

Trained in and later contracted for academic AI application technical projects. Built scalable text processing pipelines and supervised ML model development.

Research Scientist

Amsterdam UMC & CIC BioGUNE | Jan 2020-Dec 2021

Directed breakthrough biomedical research leading to novel therapeutic discovery, published in Nature Communications. Led technical projects from hypothesis to production-ready systems.

EDUCATION

MSc Applied Data Science & AI | Open Institute of Technology | 2025-26 (Expected)

MSc Molecular Biology & Biomedicine | University of the Basque Country | 2019-2020 | 9.15/10.0

BSc(Hons) Combined STEM | The Open University, UK | 2018-2019 | First Class Honours

Engineering Sciences (Bioengineering) | Dartmouth College, USA | 2013-2017 | 193 ECTS credits

KEY ACHIEVEMENTS

- Technical Innovation: Developed novel automation architectures and complex orchestrations reducing manual processing time by ~60% across multiple client projects
- Perfect Client Delivery: 100% satisfaction rating on Upwork across 23+ technical projects
- Novel Cancer Therapeutic Discovery: Led research published in Nature Communications establishing new immunotherapy target for cancer treatment